AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL **Submission:** model_7

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{7}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

A complete, correct solution. It proves the common residue modulo g and that $c^{k-1} \equiv -1 \pmod{g}$ with the residue invertible, bounds $g < 500000$ for $k \geq 4$, handles the $k = 3$ case via $g = (r^2+1)/m$ with $m \in \{1, 2, 3\}$ (correctly eliminating $m = 3$ on divisibility grounds), and verifies the 998285 construction. No errors. Scores **7**.